THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS FULL SYLLABUS PAPER SET 08

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The distance between $A(2,\ -1)$ and $B(10,\ 14)$ is: 1

(a) 17 units **(b)** 19 units **(c)** 18 units **(d)** 16 units

2. A student says: "The points $(2,\ 3)$ and $(3,\ 2)$ are the same point." The correct response is: 1

(a) False; they are different because $(x, y) \neq (y, x)$ when $x \neq y$

(b) True if they are on the same line

(c) True only in Quadrant I

(d) True

3. The point $(-5,\ 0)$ lies: 1

(a) Quadrant III **(b)** Quadrant I

(c) on the x-axis **(d)** Quadrant II

4. The degree of the polynomial $4z - 3$ is: 1

(a) 1 **(b)** 2 **(c)** 0 **(d)** 3

5. The point $(2,\ 5)$ ___ the line $y = 3x - 2$. 1

(a) is the origin **(b)** does not lie on the line

(c) is the y-intercept of the line **(d)** lies on the line

6. A student claims: "$\sqrt{2} \times \sqrt{2}$ is irrational, as both numbers are irrational." The correct response is: 1

(a) False; $\sqrt{2} \times \sqrt{2} = 2$ is rational **(b)** True only for square roots

(c) True **(d)** Cannot be determined

7. The number $1.01001000100001\ldots$ is: 1

(a) irrational **(b)** not a real number

(c) neither rational nor irrational **(d)** rational

8. A rational number in lowest terms has denominator $2^3 \times 5$. Its decimal expansion terminates after how many places? 1

(a) 2 **(b)** 4 **(c)** 5 **(d)** 3

9. $(2n + 7)(2n - 7)$ equals: 1

(a) $4n^2 - 45$ **(b)** $4n^2 - 28n + 49$ **(c)** $4n^2 + 28n + 49$ **(d)** $4n^2 - 49$

10. The coefficient of ab in $(a + b + c)^2$ is: 1

(a) 3 **(b)** 2 **(c)** 4 **(d)** 1

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11. An arc subtends an angle of 100° at the centre. The angle it subtends at a point on the remaining part of the circle is: 1

(a) 50° **(b)** 200° **(c)** 100° **(d)** 130°

12. A chord of length 18 cm is at a distance of 12 cm from the centre of a circle. The radius of the circle is: 1

(a) 15 cm **(b)** 18 cm **(c)** 16 cm **(d)** 21 cm

13. A chord of length 14 cm is drawn in a circle of radius 25 cm. The distance of the chord from the centre is: 1

(a) 32 cm **(b)** 8 cm **(c)** 18 cm **(d)** 24 cm

14. The length of a semicircle of radius r is: 1

(a) $2\pi r$ **(b)** πr^2 **(c)** πr **(d)** $\frac{\pi r}{2}$

15. By Heron's formula, the area of a triangle with sides 13 cm, 14 cm and 15 cm is: 1

(a) 182 cm^2 **(b)** 78 cm^2 **(c)** 84 cm^2 **(d)** 90 cm^2

16. Taking $\pi = \frac{22}{7}$, the circumference of a circle of radius 14 cm is: 1

(a) 616 cm **(b)** 88 cm **(c)** 44 cm **(d)** 176 cm

17. The number of outcomes in the sample space when two coins are tossed together is: 1

(a) 2 **(b)** 4 **(c)** 6 **(d)** 8

18. As the number of trials increases, the experimental probability tends to get closer to the: 1

(a) sample space size **(b)** theoretical probability
(c) value 1 **(d)** number of trials

19. If $T_1 = 1, T_2 = 2, T_3 = 4$ and $T_n = T_{n-1} + T_{n-2} + T_{n-3}$, then T_5 is: 1

(a) 11 **(b)** 14 **(c)** 13 **(d)** 12

20. The 12th term of the AP $-5, -1, 3, \ldots$ is: 1

(a) 40 **(b)** 78 **(c)** 39 **(d)** 38

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. Find the coordinates of the midpoint of the segment joining $P(0,\ 0)$ and $Q(-8,\ 10)$. 2

22. Write the number of tiles in stage n if the pattern $1, 3, 5, 7, \ldots$ is followed, and find which stage has 47 tiles. 2

23. Without long division, state whether $\frac{7}{12}$ has a terminating or a repeating decimal. Give a reason. 2

24. Factorise $9x^2 + 24xy + 16y^2$. 2

25. ABCD is a cyclic quadrilateral with $\angle A = 3x^\circ$ and $\angle C = (2x + 30)^\circ$. Find x and $\angle A$. 2

26. Find the length of the arc of a circle of radius 3.5 cm if the angle at the centre is 60° ($\pi = \frac{22}{7}$). 2

27. Two dice are rolled together. Find the probability that the sum is 7. 2

28. Is $1, -1, 1, -1, 1, \ldots$ a GP? Give the common ratio and the nth term. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. Find the perimeter of the triangle with vertices $(0,\,0)$, $(4,\,0)$ and $(4,\,3)$. 3

30. The wire of length 20 cm is bent into a rectangle of length x cm. Write the area in terms of x and find it for $x = 6$. 3

31. Show that the average $\frac{a+b}{2}$ of two rational numbers $a < b$ lies between them. 3

32. Factorise $6x^2 + 7x + 2$ by splitting the middle term. 3

33. A circular park has perimeter 88 m. Find its area and the cost of fencing it at ₹12 per metre ($\pi = \frac{22}{7}$). 3

34. A spinner has 8 equal sectors numbered 1 to 8. Find the probability that it stops on 3
(i) a prime
(ii) a number greater than 5.

35. ABCD is a cyclic quadrilateral with $\angle ABC = 110^\circ$. Find the reflex angle AOC at the centre O, where the arc is ADC. 3

OR

Prove that equal chords of a circle are equidistant from the centre. 3

36. Verify that $1 + 3 + 5 + 7 + 9 = 5^2$ and prove that the sum of the first n odd numbers is n^2 using the AP sum. 3

OR

Check whether 308 and 471 are terms of $s_n = 5n - 2$. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** Brahmagupta's rules: fortune + fortune = fortune; debt + debt = debt; debt × fortune = debt; debt × debt = fortune. Using these, 5
(i) compute $(-3) \times 4$ and $(-3) \times (-4)$;
(ii) a spice trader borrows ₹850, earns ₹1,200 and loses ₹450 — find his final standing;
(iii) explain $(-3) \times (-4) = 12$ in terms of debt.

OR

Convert into $\frac{p}{q}$: 5
(i) 12.6
(ii) 0.0120 (terminating)
(iii) 3.052 (terminating)
(iv) $0.\overline{23}$
(v) $2.0\overline{5}$.

38. Prove that if two chords of a circle are equidistant from the centre, they are equal; then find the length of each chord if the distance is 12 cm and the radius is 13 cm. 5

OR

(i) Prove that an angle inscribed in a semicircle is 90°. 5
(ii) Chord AB subtends $\angle ACB = 62^\circ$ at C on the major arc. Find the angle AB subtends at the centre and at a point D on the minor arc.

39. (i) Prove that the two paths from P to Q — one a semicircle and the other made of n semicircles on adjacent parts of PQ — have equal length. **5**

(ii) Hence find the length of each path if $PQ = 28$ cm and the smaller semicircles have diameters 7, 7, 14 cm. ($\pi = \frac{22}{7}$)

OR

Baudhāyana's problem: **5**

(i) Explain what it means to ‘square a rectangle’.

(ii) Find the side of the square equal in area to a $25\ \text{cm} \times 4\ \text{cm}$ rectangle.

(iii) Find its perimeter and compare it with the rectangle's perimeter.

40. (i) Prove that $t_n + t_{n-1} = n^2$ for triangular numbers. **5**

(ii) Verify for $n = 10$.

(iii) Find the 17th and 80th triangular numbers.

OR

Classify each as AP, GP or neither, and give the nth term where it exists: **5**

(a) $7, 11, 15, 19, \ldots$

(b) $2, 6, 18, 54, \ldots$

(c) $1, 4, 9, 16, \ldots$

(d) $100, 50, 25, 12.5, \ldots$

END OF QUESTION PAPER **Total: 80 Marks**

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